Fall 2025

Statics Mth304

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Total Marks: 20

Due date: 5th November, 2025

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Question 1:

An 20N force acts on a 3kg box, which is i
Determine the contact force between them.

Given:

F=20N

Mass of box 1:= 3kg

Mass of box 2:= 5kg

smooth surface =0

Find the acceleration of the system

Both boxes move together since they are in contact.

$$a = \frac{F}{m_1 + m_2}$$

$$a = \frac{20}{3+5}$$

$$a = \frac{20}{8} = 2.5 m/s^2$$

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Find the contact force between the two boxes

Consider the **5 kg box** alone.

The only horizontal force acting on it is the **contact force (Fc)** from the 3 kg box.

Using Newton's 2nd law:

$$F_c = m_2 \times a$$

$$F_c = 5 \times 2.5$$

$$F_c = 12.5N$$

Question 2: **Marks = 05**

Two forces act on a point such that one force is twice the magnitude of the other. If their resultant is perpendicular to the smaller force. What would be the angle between the two forces?

Given:

Let

- Smaller force = F
- Larger force = 2F
- The resultant R is **perpendicular** to So, R⊥F

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We must find the **angle θ** between the two forces.

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Step 1: Use vector addition formula for the resultant

$R^2 = F^2 + (2F)^2 + 2(F)(2\,\mathrm{F})\cos\theta$

$R^2 = F^2(1+4+4\cos\theta)$

$R^2 = F^2(5+4\cos\theta)$

Step 2: Since R⊥F, the dot product of R and F is zero.

That means the component of 2F along F must cancel with part of F itself.

From the **parallelogram law**, the component of the resultant along F is:

$R\cos\phi = F + 2F\cos\theta$

But since R⊥F

$R\cos\phi = 0 \Rightarrow F + 2F\cos\theta = 0$

Solve for cosθ

$1 + 2\cos\theta = 0$

$\cos\theta = -1/2$

Find θ

$\theta = 120^\circ$

Question 3:

Marks = 05

Find the magnitudes of two forces if their resultant is $\sqrt{5}$ N when they act at right angles, and $\sqrt{7}$ N when they act at an angle of 60°.

Given:

Let the two forces be P and Q.

When they act **at right angles (90°):**

$$\underline{R_1 = \sqrt{P^2 + Q^2} = \sqrt{5}}$$

$$P^2 + Q^2 = 5 \rightarrow (1)$$

$$at\ 60^0 :$$

$$R_2 = \sqrt{P^2 + Q^2 + 2PQ\cos 60^0} = \sqrt{7}$$

$$P^2 + Q^2 + 2PQ(\frac{1}{2}) = 7$$

$$P^2 + Q^2 + PQ = 7 \rightarrow (2)$$

$$\text{Subtract}\ (1)\ from\ (2)$$

$$P^2 + Q^2 + PQ - P^2 + Q^2 = 7 - 5$$

$$PQ = 2$$

$$x^2 - 3x + 2 = 0$$

Question 4: **Marks = 05**

Tensions be T_1 (at 30 degree to the horizontal) and T_2 at (45 degree to the horizontal) weight W=120N acts downward.

Equilibrium given two equation:

1. **Horizontal components cancel:**

   $T_1 \cos 30^o = T_2 \cos 45^0$

   so

   $$T_1 = \frac{T_2 \cos 45^0}{\cos 30^o} = T_2\sqrt{\frac{2}{3}}$$

2. **Vertical components sum to the weight:**

$$T_1 \sin 30^o + T_2 \sin 45^0 = 120.$$

so

$$T_1 = \frac{T_2 \cos 45^0}{\cos 30^o} = T_2 \sqrt{\frac{2}{3}}$$ 03255617683

Substitute T_1

$$T_2 \sqrt{\frac{2}{3}} . \frac{1}{2} + T_2 \frac{\sqrt{2}}{2} = 120$$

$$T_2 = \frac{-240}{\sqrt{\frac{2}{3}} + \sqrt{2}} \approx 107.6N$$

$$T_1 = T_2 \sqrt{\frac{2}{3}} \approx 87.8N$$

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